

Bias and Variance – Exam Notes

1. Bias

Definition:

Bias is the error due to overly simple assumptions in the model.

Key Idea:

Model does not learn the true pattern of data.

Characteristics:

- High bias → model is too simple
- Leads to **underfitting**
- Misses important relationships

Example:

Using a linear model for complex (non-linear) data.

2. Variance

Definition:

Variance is the error due to too much sensitivity to training data.

Key Idea:

Model changes a lot with small changes in data.

Characteristics:

- High variance → model is too complex
- Leads to **overfitting**
- Captures noise along with pattern

Example:

Very deep decision tree that memorizes data.

4. Bias-Variance Tradeoff

Definition:

It is the balance between bias and variance to build a good model.

If bias increases → variance decreases

If variance increases → bias decreases

5. Relationship Summary

- **High Bias + Low Variance → Underfitting**
- **Low Bias + High Variance → Overfitting**
- **Balanced Bias & Variance → Good Model**